Prerequisite-Aware Study Scheduling

Exact and Approximation Algorithms for DAG-Constrained Knapsack

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Course: Introduction to Algorithms (SL 203)

1. Problem Statement

Given:

- A set of chapters $i = 1, 2, \ldots, n$
- Each chapter has:
  - Study time t_i
  - Marks (value) m_i
  - A set of prerequisite chapters
- Total available time budget W

Objective:

- Select a subset of chapters such that:
  - Total time $\leq W$
  - All prerequisite constraints are satisfied
- Maximize total marks:

$$\max \sum m_i$$

The Core Difficulty

This is **not** a standard knapsack. here, item selection is constrained by a dependency graph. Choosing a high-value item forces you to also "pay" for all its prerequisites.

2. Problem Structure: Dependency DAG

Key Challenge:

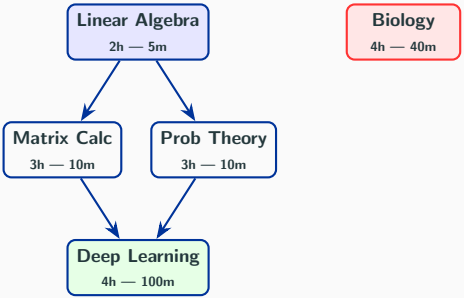
- Chapters are **NOT independent**
- Dependencies form a **Directed Acyclic Graph (DAG)**
- A chapter can be chosen **only if all its prerequisites are completed**

The Scheduling Dilemma

High Value Leaf: "Deep Learning" (100m) requires 3 foundational math nodes.

Easy Trap: "Biology" (40m) is independent and cheap.

A greedy or standard DP solver picks Biology and loses the keys to Deep Learning.



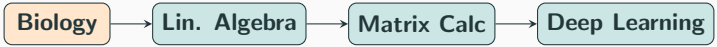
Budget $W = 10h$. {Linear Algebra, Matrix Calc, Prob theory, Deep Learning} $= 12h > W$ (Impossible without prerequisite planning).

3. Topological Sort: The Processing Backbone

For every prerequisite edge $u \to v$, chapter u appears before chapter v.

Dependency graph:

Valid topological order:



Both Biology and Lin. Algebra have no prerequisites, either can go first.

3. Why Normal DP Fails: The "State Memory" Problem

Standard DP (Bellman): $dp[w] = \max(dp[w], dp[w-t] + m)$

- **Hidden assumption:** Items are independent.
- At $W = 5$ hours, DP sees:
  - **Option A:** {Linear Algebra + Matrix Calc} = 15m
  - **Option B:** {Biology} = 40m
- **The Bug:** DP **evicts** Option A because $40 > 15$.
- **Consequence:** The "Keys" to future high-value nodes are deleted from the table forever.

Budget	Selection	Status
5h	{Biology}	Winner
5h	{Math Path}	Evicted!

The Core Flaw

Standard DP assumes all items are "Terminal." It does not know that a low-value node might be a mandatory prerequisite for a global optimum.

4. The SAT-Gate: A Lock-and-Key Fix

Core Idea: A chapter can be selected **ONLY if all prerequisites are already selected.**

Bitmask Transition Rule:

A node v can be added to state (w, mask) iff:

$$(\text{mask} \ \& \ \text{prereq_mask_v}) = \text{prereq_mask_v}$$

- **Analogy:** Each chapter = a locked door; prerequisites = keys. Without the keys you cannot open the door.
- **Fix:** Add dependency check before every DP transition.

Memory-Aware: The Algebra path and the Biology path coexist at the same weight because their bitmasks differ. hence, no eviction of prerequisite chains.

5. SAT-Gated Greedy

Idea: Select chapters by efficiency $\frac{m_i}{t_i}$, but only if prerequisites are satisfied.

Method:

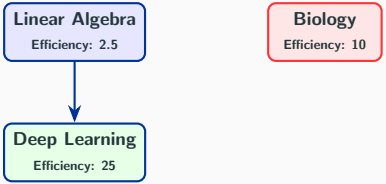
1. Sort chapters by $\frac{\text{marks}}{\text{time}}$ (descending)
2. For each chapter, select if:
   - Fits within remaining budget
   - All prerequisites already selected

Advantage

Fast and simple: $O(N \log N)$ runtime. Good for quick initial estimates over large syllabi.

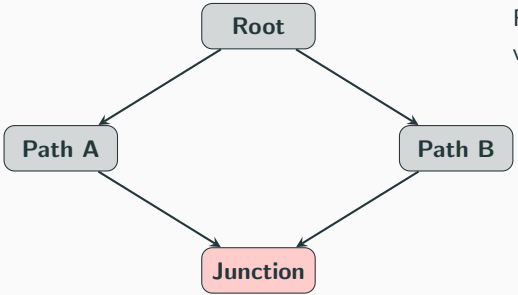
Limitation

Makes **local decisions** only, cannot see future benefit chains. May reject a low-value prerequisite that would unlock a very high-value leaf node.



Greedy picks Biology (10 greater than 2.5), missing the DL chain.

6. SAT-Gated DP: The Diamond Problem



Root is a shared ancestor of **Junction** via two independent paths.

The Core Failure

Root's time gets **counted twice**-once via Path A, once via Path B.

- Junction's prerequisite check *passes*-Path A and Path B are both selected
- But the 1D `dp[w]` table has **no memory** of which items built each state
- **No fix exists** within this framework

7. Bitmask SAT-Gated DP: Correctness vs. Scale

Idea: Store a **bitmask history** of the bag: enumerate all 2^N subsets and for each check prerequisites and the time constraint.

Guarantee

Always finds the optimal solution. Mathematically exact. Handles diamonds, chains, and any DAG topology.

The Exponential Barrier

Complexity is $O(2^N \cdot N)$. For 1,000 topics this requires 2^{1000} states. Completely infeasible.

N Topics	Memory	Outcome
10	~4 KB	Instant
20	~800 MB	Feasible
30	~800 GB	**RAM Wall**
5000	∞	**Impossible**

Solution Path: We must move from indexing by **Weight** to indexing by **Scaled Value**. This is FPTAS.

8. FPTAS: Polynomial-Time Approximation

Core Insight: We don't need perfect precision for massive datasets. We scale the "Marks" axis.

$$K = \frac{\epsilon \cdot M_{max}}{N} \implies \text{New Marks: } m'_i = \lfloor m_i / K \rfloor$$

- **Speed:** Complexity drops from $O(2^N.N)$ to $O(N^2/\epsilon)$.
- **Precision:** We guarantee a solution $\geq (1-\epsilon) \times OPT$.
- **Intuition:** Small loss in precision $\rightarrow$ huge gain in efficiency.

Transition to Scale

At $N = 5{,}000$, FPTAS discretizes the value-space. This allows us to handle the full university syllabus within milliseconds while maintaining rigorous mathematical bounds.

9. Final Comparative Benchmarking

Criterion	Greedy	SAT-Gated DP	Bitmask DP	FPTAS
Correctness	Heuristic	Exact (DAGs)	Mathematically Exact	$\geq (1-\epsilon)\cdot\text{OPT}$
Time	$O(N \log N)$	$O(N \cdot W)$	$O(2^N.N)$	$O(N^2/\epsilon)$
Space	Low	High	Very-High (RAM)	Medium
Best Scenario	Initial est.	Medium DAGs	Small exam ($N < 20$)	Full syllabus (N=5000)

Our Contribution:

We provide a **hybrid pipeline**: SAT-Gated Greedy for fast estimates, **Bitmask DP** for theoretical validation on small inputs, and **FPTAS** for real-world academic scale, each layer provably correct within its operating range.

10. Future Scope

Multi-Constraint Knapsack

Extend to joint budgets of time and cognitive load to model realistic student fatigue.

Adaptive Scheduling

Re-optimize the study plan in real time as new chapters or prerequisites are added.

Personalized Weights

Learn per-student time and marks estimates from past performance to tailor recommendations.

Long-Term Vision

A near-optimal, personalized study planner integrated with university LMS systems, treating curriculum planning as a principled optimization problem.

Thank You

Questions?